THESIS PROGRESS · ESS IN DAY-AHEAD MARKETS

Storage bids, state of charge and market monitoring

Reaching the efficient storage outcome without redesigning the market, and what it is useful for

1 **Efficient design**
exists, but is hard to adopt

2 **Today's bids**
can reach the same outcome

3 **Two uses**
value of reform · market monitoring

Carlos · October 2026

Who manages the battery's state of charge?

THE PROBLEM

Today: bid-based design (e.g. ERCOT)

- The battery submits a charge bid and a discharge offer for every hour.
- The market does not see its state of charge (SOC).
- The owner must anticipate prices to bid well, and errors cost efficiency.

Proposal: SOC-based design (Baldick, 2026)

- The battery submits its technical data and a value for its final SOC.
- The market operator (ISO) optimises the SOC with the rest of the system.
- No price forecasts needed: the efficient benchmark.

Same batteries, same system, two ways of scheduling them. The question is how far apart their outcomes are, and why.

Why it matters Batteries increasingly set prices, so how they are scheduled affects costs and prices for everyone.

Why not simply adopt the SOC-based design?

THE OBSTACLE

Costly, slow reform

New rules, software and approval. Baldick: it “would require considerable changes” to existing markets.

Responsibility moves to the ISO

The ISO would run the batteries. ISOs don't want that liability; owners lose control of their own equipment.

Strategy moves, it doesn't vanish

Batteries could misreport technical data; Baldick already foresees mitigating them.

Heavier computation

Extra size of the ISO problem (E batteries, T intervals; nnz = nonzero entries of the matrix):

$$\Delta\text{rows} = 3ET, \qquad \Delta\text{cols} = ET, \qquad \Delta\text{nnz} = 8ET - 3E$$

SOC chains every interval to all earlier ones, so the problem can't be split by interval:

$$s_{e,t} = \gamma_e^t s_{e,0} + \Delta T \sum_{\tau=1}^{t} \gamma_e^{t-\tau} \left(\alpha_e p_{e,\tau} - \beta_e g_{e,\tau}\right)$$

Solve time, SOC design ÷ current design

intervals/day × batteries	24 (1 h) × 1	96 (15 min) × 200	288 (5 min) × 200
slower by	1.6×	13×	**112×**

Same one-day horizon at finer resolution; Baldick's example scaled up (LP, HiGHS). Feasible in practice (MISO, >100 batteries), but after the market it is no problem at all: ideal as an ex post benchmark.

Why it matters Markets will keep interval bids for years. We need to know how close those bids can get to the efficient outcome.

OUR QUESTION

Can today's bids reproduce the efficient outcome, and what information do they need?

Yes

with the right bids, same dispatch and same prices

Data + prices

the bids follow from the battery's data and its local prices

Limits

information only matters when the SOC hits a limit

The efficient outcome can be reproduced with interval bids

RESULT 1 · EQUIVALENCE

Baldick's clearing
efficient dispatch and prices

→

Value of stored energy
π, and SOC rents v

→

Interval bids
charge bid u, discharge offer c

→

Current design
same dispatch, same prices

The bids

$$u^{p\star}_{e,t} = \alpha_e\left(\pi^{\star}_{e,t} - \bar{\nu}^{\star}_{e,t}\right) - \epsilon_e, \quad \hat{p}_{e,t} = \bar{p}_e \qquad c^{g\star}_{e,t} = \zeta_e + \beta_e\left(\pi^{\star}_{e,t} + \underline{\nu}^{\star}_{e,t}\right), \quad \hat{g}_{e,t} = \bar{g}_e$$

- Offer = degradation cost + value of the energy it gives up (π, plus a rent if it empties).
- Bid = value of the energy it stores (π, minus a rent if it fills), net of degradation.
- Proof: both designs then share exactly the same optimality (KKT) conditions.

Why it matters The two designs are not fundamentally different: the gap between them is information, not market rules.

Anyone with the battery's data and a price path can compute the bids

RESULT 2 · ONLY DATA AND PRICES

Battery data

efficiencies, losses, capacity, power, final-SOC value

→

Prices at its node

one price per hour

→

Price-taking schedule

the battery's own optimisation (an LP)

→

Bids

charge bid and discharge offer per hour

What it does not need

- Other participants' offers
- The network and its constraints
- The ISO's solver or its dual variables

Formally (Theorem 1)

Bids = Φ(battery data, prices). Any optimal dual of the battery's own price-taking schedule gives bids that reproduce the efficient outcome. All admissible bids form a simple set (Proposition 5).

Why it matters It turns a market-design question into a computable tool: the battery owner or the ISO can evaluate it.

Prices set the value of stored energy, and the value sets the bids

HOW IT WORKS

Baldick's base case	t = 1	t = 2	t = 3	t = 4	t = 5	t = 6
Price ($/MWh)	43.1	50	0	0	50	50
Battery	charge	**discharge, empties**	charge	**charge, fills**	discharge	**discharge, empties**
Value of 1 MWh stored, π	**49**	**1.11**	**1.11**	**49**	**49**	**40**
Offer c = 1 + max{π start, π end}	50	50	2.11	50	50	50
Bid u = 0.9·min{π start, π end} − 1	43.1	0	0	0	43.1	35

1 Between limits

the battery buys below a threshold and sells above one; the most expensive hour it still buys in sets the value

2 At a limit

the value jumps (a rent): energy cannot be carried past an empty or full battery

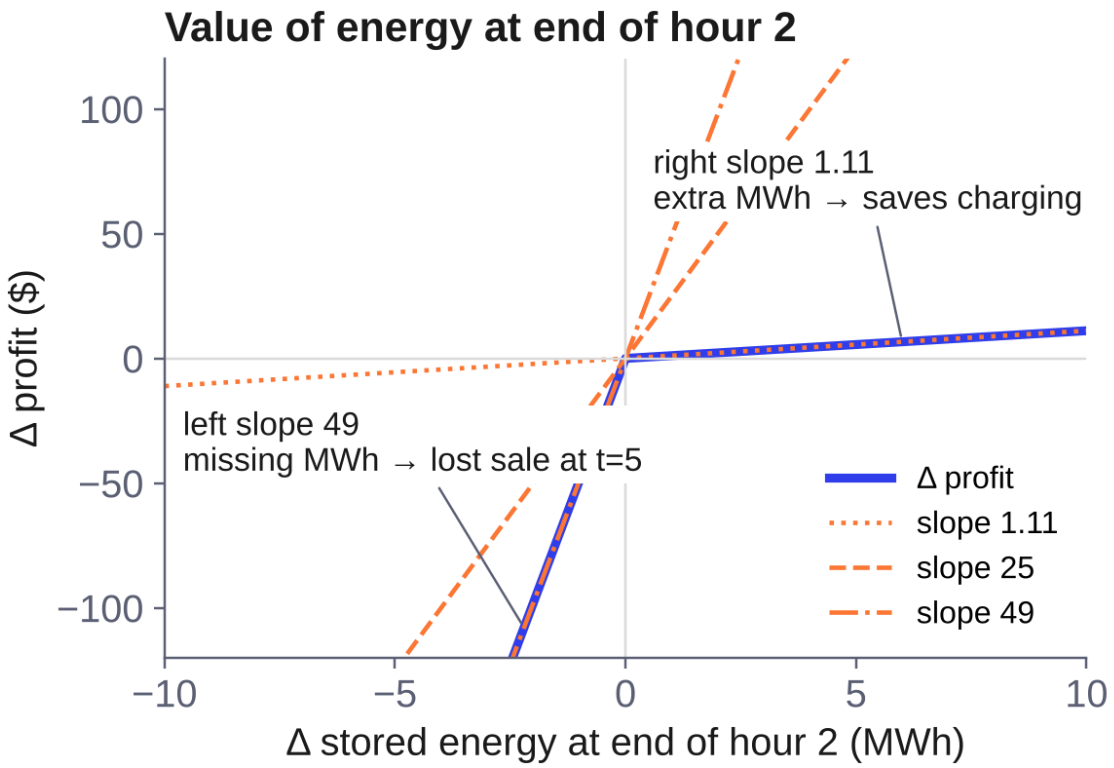
3 The bid of an hour

compares the value at its start and its end: narrow band without a limit, wide band at a limit

Why it matters The bids are simple and explainable: a few prices and the battery's own data are enough.

When the battery hits a limit, prices give a range, not a number

WHERE INFORMATION MATTERS



Base case, end of hour 2 (battery empty)

- One extra MWh only saves cheap charging: worth 1.11.
- One missing MWh costs a sale at 50: worth 49.
- Any value in [1.11, 49] reproduces the same prices.
- But each choice protects against a different forecast error, so we need a rule to choose (the pre-limit rule).

Why it matters This is exactly where forecast errors hurt the current design, and where a reform or a better rule pays off.

Measuring what the reform is worth

USE 1 · FOR REGULATORS

Welfare gap

$$\Delta SW = SW^{SOC} - SW^{current}$$

- Baldick's model, run ex post, gives the efficient benchmark.
- The current design is run with bids built from price forecasts.
- Perfect forecast: $\Delta SW = 0$ (our theorem). Imperfect: $\Delta SW > 0$.

Small gap

the reform may not pay off: well-built bids are enough

Large gap

a quantitative argument for the SOC-based design

Why it matters It gives regulators a number: how much welfare is lost by keeping today's market design.

A competitive reference bid to monitor batteries

USE 2 · FOR MARKET OPERATORS

CAISO today

- If a battery may exercise market power, CAISO can replace its bid with a default energy bid.
- Its opportunity cost uses one price, e.g. the 4th-highest day-ahead price for a 4-hour battery, plus a 10% adder.

With our result

- The ISO computes Φ(battery data, its own price forecast): an hour-by-hour, SOC-consistent reference bid.
- No change to the clearing engine is needed.

Screening

compare submitted bids with the reference

Mitigation

replace bids that fail a market power test

Tolerance band

the admissible range at SOC limits: bids inside are competitive

Why it matters It gives ISOs a more exact tool to detect battery market power, without redesigning the market.

Sources: CAISO DMM, Battery bid cost recovery and mitigation data (2024); CAISO, Limited ESR opportunity cost (2026).

Where the thesis stands

RESEARCH QUESTIONS

RQ1

Can interval bids reproduce the efficient outcome, and with what information?

Answered: equivalence note (Theorems and Propositions 1–5)

RQ2

How much welfare is lost when bids use imperfect price forecasts?

Next: experiments on Baldick's cases and a simplified RTS-GMLC system

RQ3

Can ISO reference bids separate competitive from strategic batteries?

Proposed extension: same models plus strategic bidding scenarios

Why it matters One model answers three audiences: market designers, regulators and market monitors.

THE STORY IN ONE MINUTE

From an efficient design to useful tools

1. The efficient storage design needs a costly reform and moves responsibility to the ISO, so markets keep interval bids.

2. We prove that interval bids can reach the efficient outcome, using only the battery's data and its local prices.

3. So the gap between designs is information; measuring it tells regulators what the reform is worth.

4. The same function gives ISOs an exact, SOC-consistent reference bid to monitor battery market power.

Reference: R. Baldick, "A state-of-charge based formulation for storage participation in electricity markets," arXiv:2606.14993, 2026.


DTU